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Accent-Based Evaluation of Speech Anti-spoofing Countermeasures Across Multiple Languages

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Universidad de los Andes

[SPEAK IN SPANISH · DO NOT TRANSLATE YET]

Your mother calls.
She needs
$1,500, now.
It is her voice.

— and it is not her.

If your voice is your password, then every voice note you have ever sent is your password, published.

Voice biometrics authenticate banking customers by phone today, at scale. Behind that authentication sits an automatic speaker verification system, and in front of it, a countermeasure trained to spot fakes.

That countermeasure is what this paper is about.

Three on the record. Two of them defeated a machine.

2023

A journalist cloned his own voice and used it to pass his bank's Voice ID and reach his balance.

voice authentication bypassed
Vice · Feb 2023

2024

A finance worker in Hong Kong joined a video call where every other participant was synthetic, and transferred the money.

US$ 25 million
CNN · Arup, May 2024

2024

Robocalls carrying a synthetic voice of a sitting US president went out before a primary election.

US$ 1 M settlement + US$ 6 M proposed
FCC enforcement · 2024

The first one is this paper's subject. Someone spoke, a machine decided it was him, and it was wrong.

Vice, "How I Broke Into a Bank Account With an AI-Generated Voice" (2023) · CNN, Arup deepfake scam (2024) · FCC, Lingo Telecom settlement and Kramer NAL (2024).

Which of these two sounds like a real person?

A

[embed clip here]



B

[embed clip here]



Hands up for A · hands up for B

Count the hands out loud. Do not say anything else — the answer comes back in six minutes.

Universidad de los Andes

Tomás Acosta Bernal



- **AI Lead Engineer**
  Konecto — voice agents, RAG and LLM systems in production
- **Researcher**
  FLAG Lab, Universidad de los Andes — speech anti-spoofing
- **Teaching assistant**
  Natural Language Processing, Universidad de los Andes

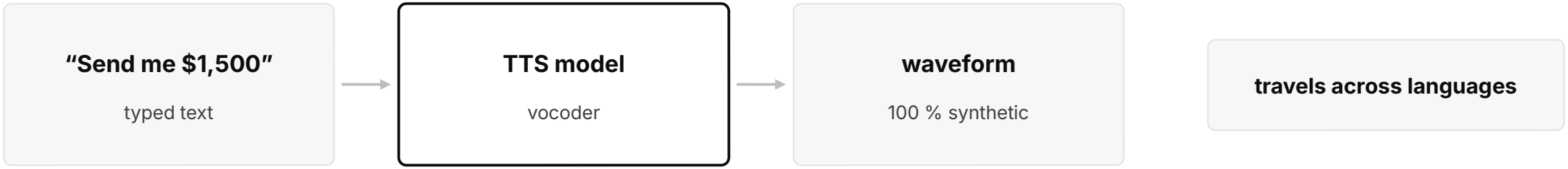
First-authored by Jaime Hurtado, with Rubén Manrique. I present it — and I build on it.

THE TWO ATTACKS

Two ways to fake a voice. Only one survives.

TEXT-TO-SPEECH

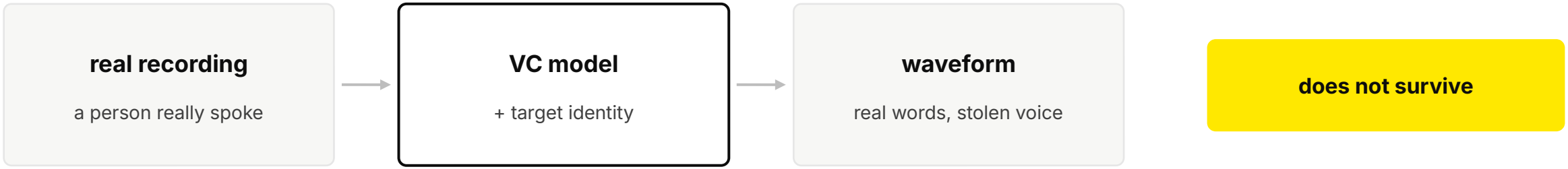
nothing real goes in



The artefact belongs to the VOCODER — it does not care what language you speak.

VOICE CONVERSION

a real recording goes in



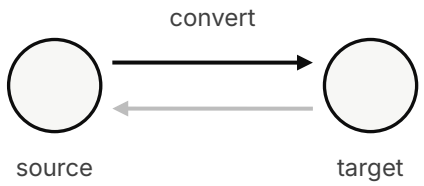
The artefact is tangled with SPEAKER and ACCENT — exactly what changes with the language.

The words are real. The identity is stolen. That is the attack our detectors cannot see in Spanish.

Three ways to steal a voice. Two GANs, one diffusion.

GAN

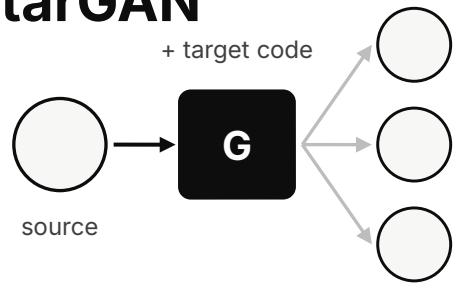
CycleGAN



Converts A into B, then back again, and checks it lands on A. That round trip lets it learn without paired recordings.

GAN

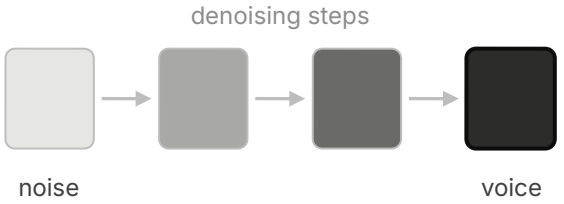
StarGAN



One generator covers many target speakers — you just tell it which voice to produce. Many-to-many.

DIFFUSION

DiffVC



Starts from pure noise and cleans it up, step by step, until a voice appears — steered by the words of the original.

And the fourth attack family: **TTS** · speech synthesised from text, not converted from a recording.

All four attacks leave a different fingerprint. That is what the detector is looking for.

Every benchmark is in English.

HABLA Spoof	2023	Spanish	logical access · deepfake
ASVspoof 2019	2019	English	logical access · physical access
ASVspoof 2021	2021	English	logical access · physical · deepfake
SAS	2016	English	logical access
FoR	2019	English	logical access
VCTK	2019	English	logical access

5 of 6

Spanish appears once. Every system in this literature learned what a fake sounds like from English speakers.

Newest entry: 2023. Multilingual corpora have since added Spanish — built from text-to-speech, without voice conversion.

Tamayo et al. (2023) showed English-trained detectors break on Spanish. We went looking for why.

Table 1 — anti-spoofing dataset taxonomy; years from the paper's own references.

What the detector hears, and how we score it.

WHAT THE DETECTOR HEARS

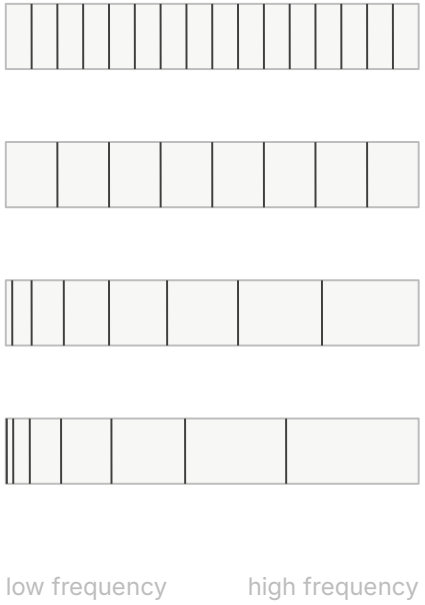
A neural network cannot eat a raw waveform. First the audio becomes a picture of energy across frequencies — and the four options differ in how they slice the frequency axis.

Spectrogram — Everything, evenly. The richest and the biggest.

LFCC — Even bands — every frequency weighted the same.

MFCC — Bands packed low, spread high — it mimics the human ear.

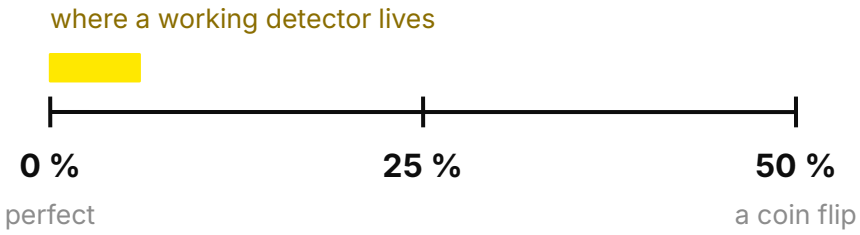
CQCC — Even more resolution down low, where synthesis leaves marks.



HOW WE SCORE IT

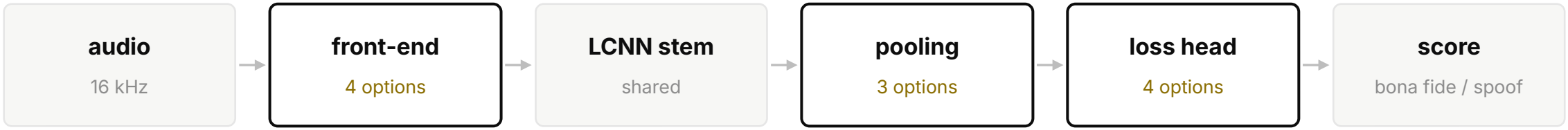
Equal Error Rate

Too strict and you reject real people.
Too loose and you let fakes through.
EER is the point where those two errors are equal — one number for the whole trade-off.



Lower is better. Every number in this talk is an EER.

One pipeline, judged six times over.



black outline = the axes we varied

THE BACK-END

A Light CNN. Max-Feature-Map activations keep the strongest channel at each position and drop the rest, so the network stays small. Every configuration shares the same stem — what changes is how the frames are collapsed into one decision:

- **attention** — weights frames by how much they matter
- **trim-pad** — forces every utterance to one fixed length
- **LSTM-sum** — reads the sequence in both directions, then sums

HOW WE JUDGED IT

Six seeds per configuration. We report medians and observed ranges, not a single best run. Architectures are compared with a Wilcoxon signed-rank test on **seed-matched pairs**, with Cohen's d and a Bonferroni correction.

With six seeds the smallest p we can reach is 0.031. The design is underpowered for small effects — so we do not crown winners we cannot defend.

864 runs, and a rule about when we are allowed to call something better.

The same grid, three times. Only the data changes.



the Spectrogram row stays bad in both Spanish conditions



EER — darker is worse. Each square is one architecture: 4 feature extractors × 3 classifier variants × 4 loss functions (AM-softmax, OC-softmax, P2S-Grad, Sigmoid).

When the training language matches the test language, everything works.

Tables 5, 7 and 8 — mean EER (%) over six seeds. Merged training is joint, not fine-tuning.

What we controlled, and what we could not.

TRAINING SET	TRAIN	DEV	TEST
English	25,380	24,844	71,237
Spanish	34,684	3,658	4,809
Combined	60,064	28,502	—

utterances · Table 4

The Spanish test partition is 4,809 utterances. A 0.1 % error rate there is a handful of trials — read those numbers as “very low”, not as precise.

✔ **CONTROLLED**

Bona fide speakers are strictly disjoint across the three splits — 29, 16 and 14, none shared. The source speakers feeding the conversion systems are disjoint too. No test speaker’s genuine voice is ever seen in training.

! **NOT CONTROLLED**

The target identities the conversion systems imitate ARE shared: all 113 test identities also appear in training. That affects 95.5 % of Spanish test utterances. Our Spanish numbers are an upper bound.

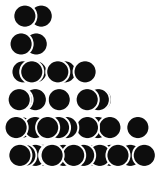
We audited the partitions ourselves, from the file lists. Saying what leaked is part of the result.

Sec. 3.4 · six seeds per configuration, early stopping on development loss, lowest-loss checkpoint retained.

Same models. We only changed the test language.

TESTED ON ENGLISH

they work



2.67 – 8.08 %

the same 48 models

TESTED ON SPANISH

they stop working

50 % = coin flip

all 48

above 30 %

best 34.4 % · 28 of the 48 above 40 %

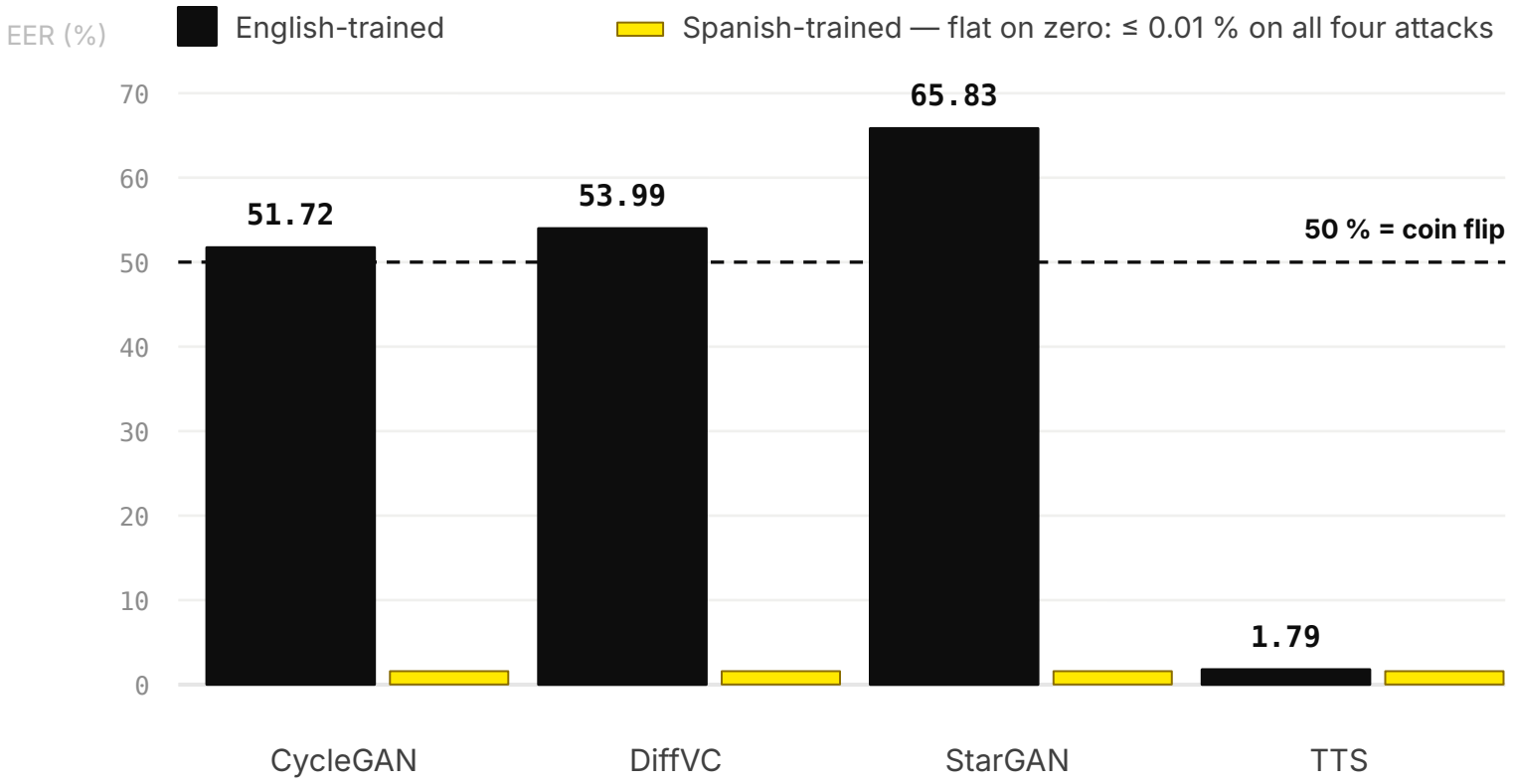


EER (%) · higher is worse

Nothing about the models changed. Only what we asked them to listen to.

Top: 48 mean EERs, Table 5. Bottom: Sec. 4.2 — all 48 exceed 30 %, lowest 34.4 %, 28 above 40 %.

The gap is not uniform. It is voice conversion.



Three tall bars = voice conversion

66 % is **worse than chance** — that model is not confused, it is systematically wrong. You would do better inverting its answer.

And the short bar is text-to-speech. Same model. Never saw a word of Spanish.

Table 6 — LCNN-LSTM-sum · P2S-Grad.

1.79 %

The same broken condition as two slides ago — English-trained, Spanish-tested — **sliced down to the synthesised attacks alone.**

Say this out loud

1.79 is the **best** English-trained model. Across back-ends the TTS column moves an order of magnitude — 1.79, 16.92, 46.11. And this is **a single TTS column** against three conversion families.

Vocoder artefacts survive the language change. Conversion artefacts do not.

Both were fake. You trusted the one the network misses.

A

TEXT-TO-SPEECH

you flagged it

1.79 %

the detector caught it too

B

VOICE CONVERSION

you trusted it

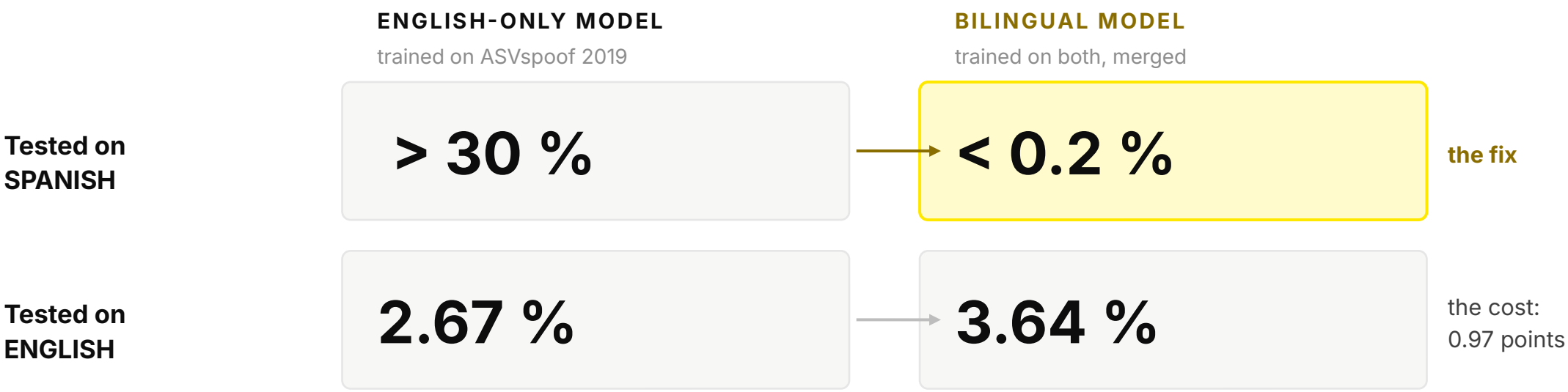
52 – 66 %

the detector missed it too

You failed at the same task, in the same direction, as a neural network trained on twenty-five thousand utterances.

EER for the best English-trained model on the Spanish evaluation set, by attack family — Table 6.

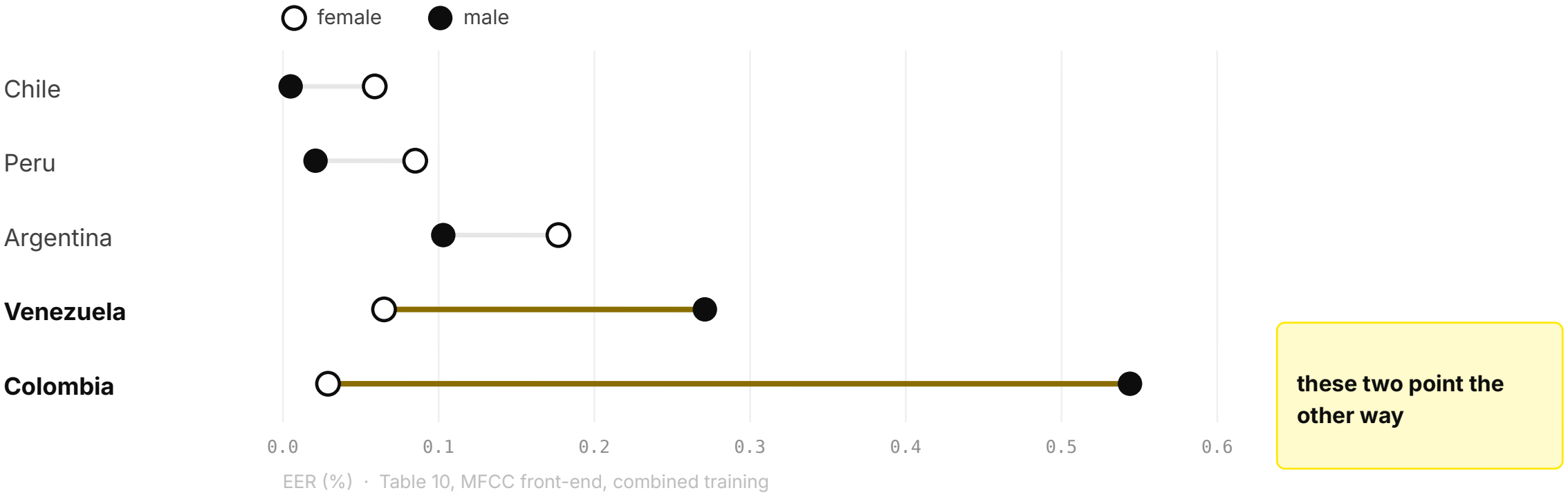
Adding Spanish fixes Spanish. The surprise is the price.



You would expect a trade-off. There is barely one — so you do not need one detector per language.

Spanish-only training reaches 0.14 % on Spanish; the paper does not test it against the bilingual model, and on a 4,809-utterance partition that difference is not one to lean on. The merged corpus is also 57.7 % Spanish, so “bilingual training helped” and “there was simply more Spanish” are not separated here. Both are named as future work.

Accent barely moves it. Gender flips direction.



Accent spread: under half a point. The gender gap is bigger — and it reverses between countries.

Colombia and Venezuela are the two that reverse, and the Colombian gap accounts for essentially all of that country's disadvantage. Because the direction is inconsistent across countries it cannot be a single bias in the front-end — the paper reads it as an interaction between speaker population and the generators used on each subset.

“Emphasise low frequencies” is only half true.

Mel-scale front-ends

TRUE

Descendant weighting gets the lowest median EER and wins 8 of 12 configurations.

Linear-scale front-ends

FALSE

The same weighting is the worst of the three scalings.

The usual justification is that the principal formants sit low in the spectrum. That holds for F1. F2 spans roughly 600–2500 Hz and does not cooperate.

Test both scalings, or you will report whichever one your front-end happened to be.

Table 11 — full configuration grid, 72 runs per scale per family.

Three things I would rather say myself.

01

Small test partition

Under 5,000 Spanish utterances. A 0.1 % error rate is a handful of trials — read it as “very low”, not as a precise number.

02

Shared target identities

The conversion attacks reuse target identities between train and test. Our Spanish numbers are an upper bound.

03

Four attack families

HABLA gives us four attack families. We just showed that conclusions depend on which attacks a corpus contains — ours carry that same dependency.

Number three is what I am working on now.

The language gap is real.
It lives in voice conversion.
One bilingual model closes it.

No architecture escapes the gap — we checked 48. And you do not need one detector per language.

THANK YOU

Questions — and what comes next

MARSA — a multi-accent Spanish corpus that adds full-spoof attacks from six contemporary TTS systems, a partial-spoof tier where individual words are spliced into bona fide utterances, and an acoustic augmentation layer. That tier needs frame-level scoring, because an utterance-level error rate has no way to express “three words of this sentence were fake”.

QR
paper · code · audio clips

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